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Quantum Meta-Reinforcement Learning for Few-Shot Anomaly Detection

MSc Research Project Report

MSc in Artificial Intelligence

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List of Abbreviations

PQC	Parameterised Quantum Circuit
ES	Evolution Strategies
ROC	Receiver Operating Characteristic (curve of true positive rate vs false positive rate)
AUC	Area Under the ROC Curve
AUC-ROC	Area under the ROC curve
F1	Harmonic mean of precision and recall
NISQ	Noisy Intermediate-Scale Quantum
SVM	Support Vector Machine
OC-SVM	One-Class Support Vector Machine
QML	Quantum Machine Learning
MLP	Multi-Layer Perceptron
RBF	Radial Basis Function (kernel)

Abstract

Anomaly detection is a critical task in domains such as healthcare, finance, and cybersecurity, where abnormal patterns may correspond to disease, fraud, or malicious activity. In practice, anomalies are rare, labelled data are scarce, and distributions drift over time, making it difficult for conventional supervised models to generalise to new domains. This project investigates a hybrid quantum classical meta-reinforcement learning framework for few-shot anomaly detection across multiple tabular datasets. A parameterised quantum circuit (PQC) encoder, trained with Evolution Strategies (ES) on episodic few-shot tasks, is combined with a classical logistic-regression head that adapts to new domains using only a small support set of labelled examples.

The system is evaluated on several source datasets spanning healthcare, radar signal data, finance, and cybersecurity, and on additional unseen OpenML datasets that serve as realistic holdout benchmarks. Cross-domain experiments show that a meta-trained PQC can substantially outperform a from-scratch PQC and strong classical baselines under a strict few-shot adaptation regime. A noise robustness study demonstrates that the PQC retains strong performance under moderate depolarising noise levels and operates within a practical hardware budget for near-term noisy quantum devices. The codebase is implemented in Python using PennyLane, Qiskit, scikit-learn, and LangGraph, with architectural optimisations such as batch-aligned ES updates, high-fidelity categorical feature preservation, adaptive threshold calibration, and fast simulation backends. Overall, the results provide empirical evidence that hybrid quantum–classical meta-learning is a viable and robust approach for few-shot anomaly detection on heterogeneous tabular data.

Keywords: Quantum machine learning, meta-reinforcement learning, few-shot anomaly detection, Evolution Strategies, parameterised quantum circuit.

1. Introduction

Anomaly detection aims to identify unusual patterns in data that do not conform to expected behaviour. In many real-world applications, anomalies correspond to events that are rare but highly consequential, such as fraudulent transactions in financial systems, network intrusions in cybersecurity, or malignant cases in medical screening. Classical anomaly detection methods often assume access to large amounts of labelled data or rely on unsupervised density or boundary estimation in feature space. In practice, labelled anomalies are scarce, the normal class may dominate the data distribution, and new domains or attack types emerge over time, making generalisation difficult [6, 7].

Few-shot learning addresses the challenge of learning from only a small number of labelled examples per class by leveraging prior experience across related tasks. Meta-learning, or "learning to learn", extends this idea by training models over a distribution of tasks so that they can rapidly adapt to new tasks with limited data. Metric-based methods such as prototypical networks learn embeddings in which classes form tight clusters, enabling classification by distance to class prototypes [1]. Optimisation-based methods such as model-agnostic meta-learning (MAML) learn initial parameters that can be quickly adapted to new tasks with a few gradient steps [2]. Practical tutorials emphasise episodic training with support and query sets to

simulate few-shot conditions [20].

Quantum machine learning (QML) explores the use of quantum computers to perform learning tasks, often via parameterised quantum circuits that act as trainable feature maps in high-dimensional Hilbert spaces [3]. Recent advances in QML software, including PennyLane and Qiskit, have made it possible to implement and differentiate hybrid quantum–classical models on both simulators and hardware [4, 5]. Survey work on quantum anomaly detection highlights promising hybrid architectures but also emphasises limitations from small synthetic datasets and hardware noise [10, 19].

This project explores a hybrid quantum–classical meta-reinforcement learning framework for few-shot anomaly detection across multiple tabular domains. A parameterised quantum circuit (PQC) is used as an encoder that maps preprocessed features into an entangled quantum state space. A classical logistic-regression head is fitted per episode on a small support set to classify anomalies in a query set. Instead of using gradient-based optimisation, the PQC parameters are trained using Evolution Strategies, a gradient-free method that can mitigate issues such as barren plateaus in quantum circuits [3]. The main goal is to learn the domain agnostic anomaly representations where it can transfer across datasets and support the rapid adaptation to new domains.

Taking these all together, the elements motivate a systematic study of hybrid quantum classical meta learning for the anomaly detection on realistic tabular datasets with an emphasis on cross domain transfer , robustness and practical deployability.

2. Related Work

2.1 Anomaly detection in finance and cybersecurity

In finance, machine-learning-based fraud detection has been studied extensively. A systematic review of studies on machine-learning-based financial fraud detection finds that support vector machines, neural networks, and ensemble methods are among the most effective approaches, while also emphasising that severe class imbalance is a dominant challenge [6]. Other work analyses financial general-ledger anomalies and reports that Isolation Forest and autoencoder-based unsupervised techniques perform well for efficiently sampling higher-risk journal entries, but that performance is sensitive to the underlying anomaly rate [7]. Isolation Forest itself, the underlying unsupervised algorithm used in both of these studies, isolates anomalies directly via random partitioning rather than profiling normal behaviour, giving it a low, near-linear time complexity that makes it attractive for large tabular datasets [9].

In cybersecurity, anomaly detection is central to intrusion and malware detection. An improved X-means and Isolation Forest methodology has been proposed for large-scale network traffic, demonstrating that combining clustering with Isolation Forest can improve anomaly ranking quality compared with several unsupervised baselines [8]. Deep learning and hybrid quantum–classical models have also been explored for network security, including quantum deep-learning-based anomaly detection methods aimed at enhancing intrusion detection [11]. These studies show that classical anomaly detection is mature and effective in many settings, but also that models are usually trained per dataset and still struggle under extreme imbalance or concept drift.

2.2 Few-shot learning and meta-learning for anomaly detection

Few-shot learning aims to learn new tasks from only a small number of labelled examples per class. Meta-learning trains a model across many tasks so that it can adapt quickly to a new one. Introductory tutorials from industry research groups provide clear descriptions of episodic training with support and query sets that simulate few-shot regimes [20].

In network anomaly detection, cross-network meta-learning approaches train a graph-based detector across several auxiliary networks and adapt it to a target network using only a handful of labelled anomalies, achieving improved performance over training from scratch on the target network alone [13]. For tabular anomaly detection more broadly, recent work argues for moving away from the conventional "one model per dataset" paradigm towards a single generalist model that is trained once across multiple source datasets and can then be applied, with little or no retraining, to previously unseen tabular domains [19]. These works confirm that handling few anomalies in tabular and network data is an active research area, but they are all purely classical.

2.3 Quantum machine learning for anomaly detection

A recent study and surveys related to quantum machine learning algorithms for anomaly detection there are various grouping methods into quantum supervised learning, unsupervised learning and reinforcement learning categories all across the domains such as cybersecurity, fraud detections and particle physics [10]. The survey emphasises that many quantum anomaly detection experiments rely on small synthetic datasets or static benchmarks and that hardware noise and resource requirements remain major constraints.

Several concrete quantum anomaly detection models have been proposed. A quantum-classical hybrid deep neural network has been designed for image anomaly detection, in which classical convolutional layers are combined with variational quantum circuit layers to learn a normality model directly from raw images [14]. Quantum deep-learning-based methods have also been developed for detecting anomalous behaviour in network traffic, with strong reported performance relative to classical baselines [11]. Closer to the one-class formulation used later in this project, quantum support vector data description (QSVDD) learns a shallow-depth quantum circuit that encloses normal data within a minimum-volume hypersphere, and has been shown to outperform quantum-autoencoder and classical deep-learning baselines on image and credit-card-fraud anomaly detection tasks [12].

In parallel, the general quantum machine learning literature has established the theoretical building blocks used throughout this project: parameterised quantum circuits as trainable machine learning models [16], the expressibility and entangling capability of such circuits [15], and the observation that hardware noise can act as an implicit regulariser rather than a purely harmful effect in shallow variational circuits [17]. However, none of the anomaly-detection-specific works above investigate episodic meta-learning of a shared PQC encoder across multiple tabular domains.

2.4 Summary and research gap

Across the literature, three themes emerge. First, classical anomaly detection is strong and widely used in finance and cybersecurity but still struggles with extreme class imbalance and domain shift [6–8]. Second, few-shot learning and meta-learning have been successfully applied to network and tabular anomaly detection with promising results, yet implementations are

exclusively classical [13, 19, 20]. Third, quantum anomaly detection has produced several promising architectures, including hybrid networks, autoencoder-style models, and one-class quantum circuits, but experiments have mostly focused on single datasets or static settings and have not explored cross-domain meta-learning [10–12, 14].

The proposed project lies at the intersection of these themes. It aims to meta-train a single PQC encoder across several tabular anomaly-detection datasets, adapt it to unseen domains with only a small support set, compare the quantum meta-learner with classical meta-learning and strong baselines, and study the effect of quantum noise and resource requirements in a way that directly addresses open questions raised in the quantum anomaly-detection literature [10, 19].

3. Research Methodology

3.1 Research questions

The project is framed by three core research questions:

- **RQ1 (Quantum vs classical encoder).** Under the same episodic meta-learning procedure, does a PQC provide better few-shot anomaly detection performance than a classical encoder on heterogeneous tabular datasets?
- **RQ2 (Cross-domain meta-learning).** Does meta-training a single PQC encoder across several datasets improve few-shot performance on an unseen dataset, compared with training the same PQC from scratch on that dataset with only a few labelled examples?
- **RQ3 (Meta-learner vs baselines).** How does the meta-trained PQC compare to strong classical baselines such as Isolation Forest and classical one-class SVM, and to a quantum one-class anomaly detection baseline, on financial fraud and intrusion detection?

In addition to these three questions, the methodology also investigates the effect of depolarising noise and hardware resource requirements, and evaluates the practical benefits of an agentic LangGraph or LangChain workflow. These are treated as methodological and evaluation aspects of the study rather than as standalone research questions.

3.2 Research design

This project follows an experimental, quantitative research design. The objective is to evaluate whether a hybrid quantum–classical framework, built around a parameterised quantum circuit encoder trained via episodic meta-reinforcement learning, can support effective few-shot anomaly detection across heterogeneous tabular datasets from healthcare, finance, cybersecurity, and remote sensing.

The study proceeds in four stages: (i) construct a unified data-processing and episodic sampling pipeline; (ii) design quantum and classical encoders and a few-shot classification head; (iii) train the encoders using Evolution Strategies (ES) across multiple source domains; and (iv) evaluate the resulting models via cross-domain holdout experiments, baseline comparisons, a noise study, and a hardware-oriented resource analysis.

3.3 Datasets

Here I have used several public tabular datasets with different source domains and unseen holdout domains. Source datasets include breast cancer diagnosis, ionosphere radar signals, credit card fraud transactions and the NSL KDD intrusion detection dataset. Unseen holdout

datasets include Diabetes, Cardiotocography and Satimage from the OpenML platform. For each dataset anomalies are defined as the minority or pathological class (for example: fraudulent transactions, network attacks or pathological fetal states).

Appendix A provides all datasets names with there links; these are treated as benchmark resources.

3.4 Data preprocessing pipeline

A shared preprocessing pipeline is applied as per dataset to produce an unified feature representation which is suitable for the both quantum and classical encoders:

1. **Train,validation and test splits**: stratified splits with preserve anomaly ratios, with fixed random seeds for reproducibility.
2. **Scaling**: the numeric features are standardised with a feature scaling method `StandardScaler` fitted on the training partition.
3. **PCA reduction**: The Principal Component Analysis (PCA) maps the features to N components where N matches the number of qubits or encoder dimensions.
4. **Categorical handling**: categorical and object-typed columns are retained by numeric casting or label encoding, avoiding information loss seen when such columns are dropped.
5. **Angle normalisation**: the PCA components are rescaled to $[-\pi, \pi]$ so that they can be directly used as angles for single-qubit rotations in the PQC. Angle encoding provides a shallow circuit and robust performance compared with basis and amplitude encoding [3].

3.5 Few-shot episode construction

Few-shot episodes are the atomic units of meta-learning. Each episode is defined on a single dataset and consists of a support set with K normal and K anomalous samples (typically $K = 5$), and a query set drawn from the remaining samples of the same dataset. During meta-training, episodes are sampled repeatedly across all source domains. For cross-domain evaluation, a dataset such as Satimage is held out from meta-training and appears only during meta-testing, where the encoder is adapted using few-shot support sets and evaluated on its query and test sets.

3.6 Experimental conditions per research question

For RQ1, a PQC encoder and a classical meta-NN encoder with similar parameter counts are trained under the same ES meta-learning protocol and evaluated on the same episodic tasks. For RQ2, the benefit of cross-domain meta-training is assessed by comparing a meta-trained PQC with a from-scratch PQC on an unseen holdout dataset using the same few-shot protocol. For RQ3, the meta-trained PQC is compared against Isolation Forest, classical one-class SVM, and a quantum one-class anomaly detection baseline inspired by prior work on classical and quantum anomaly detection [6, 7, 9, 12].

Beyond these three questions, an additional experiment evaluates the impact of depolarising noise and estimates hardware resources, following guidelines from recent analyses of quantum anomaly detection and hardware cost [10, 12]. Finally, the benefits of an agentic LangGraph/LangChain workflow are assessed qualitatively by examining reproducibility and extensibility compared with simple scripting.

3.7 Evaluation metrics

Model performance is assessed using:

- **ROC curve (Receiver Operating Characteristic).** The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) as the decision threshold is varied. Here TPR is the fraction of anomalies correctly detected (recall), and FPR is the fraction of normal points incorrectly flagged as anomalies. A curve closer to the top left corner which indicates better discrimination.
- **AUC-ROC (Area Under the ROC Curve).** The AUC is a single scalar between 0 and 1 summarising the ROC curve. An AUC of 0.5 corresponds to random guessing while 1.0 corresponds to perfect ranking and matching of anomalies above normal points. In this report, the values below 0.60 are treated as failing a minimal discrimination benchmark on difficult anomaly detection tasks.
- **F1-score.** The F1-score is the harmonic mean of precision and recall, computed at a decision threshold tuned on the support set to maximise F1. This makes F1 particularly informative under severe class imbalance.

On the OpenML holdouts, each configuration is evaluated over multiple random few-shot support splits. Mean AUC and F1 are reported, and differences are interpreted alongside their variability rather than only single best runs.

3.8 Mathematical formulation of the quantum encoder

The PQC encoder maps a classical feature vector $\mathbf{x} \in \mathbb{R}^N$ to a quantum state via angle encoding. Each feature x_i is normalised to $\tilde{x}_i \in [-\pi, \pi]$ and encoded as a rotation about the Y axis:

$$|\psi_{\mathbf{x}}\rangle = \bigotimes_{i=1}^{N} R_Y(\tilde{x}_i)\,|0\rangle^{\otimes N}\,.$$

Variational layers apply trainable rotations followed by a ring of CNOT gates:

$$U_{\text{var}}(\boldsymbol{\theta}) = \left(\prod_{i=1}^{N} R_Z(\theta_{i,z}^{(l)}) R_Y(\theta_{i,y}^{(l)})\right)\left(\prod_{i=1}^{N} \text{CNOT}(i, i{+}1 \bmod N)\right).$$

The encoder output is a vector of Pauli-Z expectations:

$$\mathbf{z}(\mathbf{x}; \boldsymbol{\theta}) = (\langle Z_1\rangle, \ldots, \langle Z_N\rangle)\,.$$

Evolution Strategies update the parameters using reward-weighted perturbations:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha \frac{1}{\sigma M} \sum_{j=1}^{M} R_j\, \boldsymbol{\epsilon}_j,$$

where $\boldsymbol{\epsilon}_j \sim \mathcal{N}(0, I)$ and R_j are normalised episode rewards (F_1-scores).

To validate quantum behaviour, standard demonstrations of Bell states and interference patterns are implemented using both PennyLane and Qiskit, confirming agreement between frame-

works [4, 5].

4. Design Specification

4.1 Overall system architecture

The anomaly-detection system has four main components, following the proposal: a data and preprocessing pipeline, quantum and classical encoders, an episodic meta-reinforcement learning loop, and an evaluation module. Data from different domains is standardised and reduced with PCA, then fed into the PQC encoder or classical encoder. The encoder outputs expectation vectors that feed into a few-shot logistic regression head trained on a small support set per episode. LangGraph orchestrates the training workflow, while LangChain can be used for optional LLM-based explanations of detected anomalies [20].

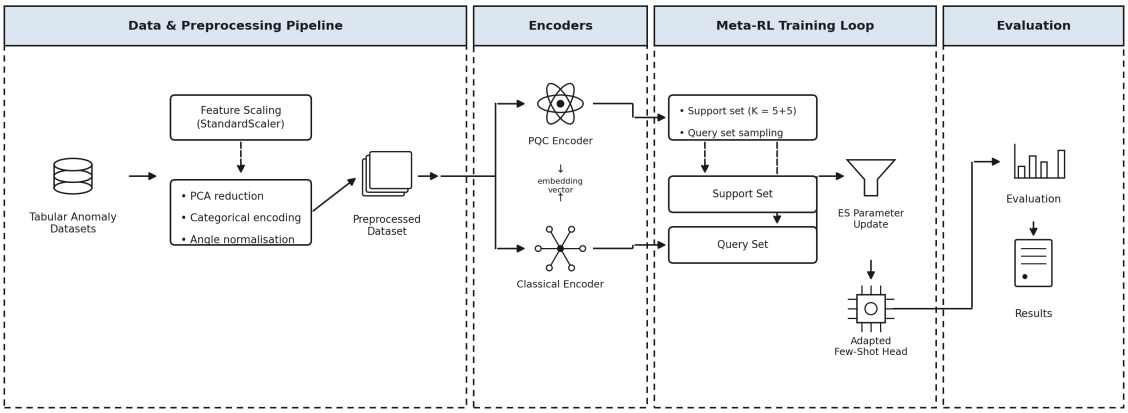


Figure 1: High-level architecture of the hybrid quantum and classical meta-reinforcement learning system for few-shot anomaly detection.

4.2 Quantum encoder (PQC)

The PQC encoder is a shallow but expressive variational circuit implemented in PennyLane [4]:

- **Qubits**: typically 4 or 8.
- **Data encoding**: angle encoding via $RY(x_i)$ rotations using normalised PCA components.
- **Variational body**: up to 3 layers of trainable RY and RZ rotations per qubit, followed by a ring of CNOT gates.
- **Measurement**: expectation values of Pauli-Z on all qubits, yielding an N-dimensional embedding vector.
- **Noise model**: optional depolarising channels to simulate NISQ hardware conditions [10, 12].

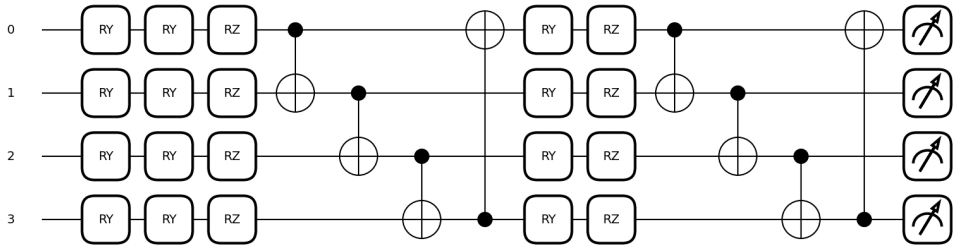


Figure 2: Example PQC encoder circuit with angle encoding, variational layers, and a CNOT ring entanglement pattern.

4.3 Classical encoder and few-shot head

The classical encoder is a small neural network with one or two hidden layers, mapping PCA features to an embedding vector of the same dimension as the PQC expectations, enabling a fair ablation under the same meta-learning loop, in the spirit of metric-based few-shot classifiers such as prototypical networks [1]. The few-shot head is typically a logistic regression classifier with class weights to mitigate imbalance, fitted on the support embeddings and used to score the query set. For each episode, the decision threshold is tuned on the support set by maximising F_1.

4.4 Meta-reinforcement learning loop

The meta-reinforcement learning loop implements episodic ES optimisation:

1. Sample a dataset and construct a support and query set.
2. Encode support and query samples through preprocessing and the encoder (PQC or classical).
3. Train the logistic head on the support set.
4. Evaluate on the query set and compute an episode reward (F_1).
5. Use ES updates over a population of perturbations to adjust encoder parameters based on aggregated rewards.

This loop is implemented as a LangGraph state machine with nodes for data loading, sampling, evaluation, parameter updates, and convergence checks, and with LangSmith logging for experiment traceability.

4.5 Baselines and quantum one-class model

Baselines reflect strong non-meta-learning competitors drawn from the literature:

- Isolation Forest, the classical isolation-based anomaly detector [9], as applied and evaluated in a financial-data setting by [7, 8],
- classical one-class SVM with RBF kernel, in line with fraud-detection practice reviewed by [6],
- a quantum one-class anomaly detector inspired by quantum support vector data description (QSVDD), which learns a minimum-volume enclosing hypersphere for normal data using a shallow variational quantum circuit [12].

These baselines provide points of comparison for credit card fraud and intrusion detection tasks.

5. Implementation

5.1 Technology stack

The implementation uses:

- `numpy`, `pandas`, and `scikit-learn` for preprocessing and classical ML.
- PennyLane for PQC construction, simulation, and automatic differentiation [4].
- Qiskit for quantum baseline circuits and cross-framework verification [5].
- LangGraph, LangChain, and LangSmith for workflow orchestration, LLM integration, and experiment logging.

5.2 Optimisation interventions

Several interventions were necessary to move from unstable baselines to reliable performance:

- **Batch-aligned ES**: evaluating all ES perturbations on the same batch of episodes reduces variance and stabilises optimisation.
- **Categorical feature preservation**: retaining categorical variables through encoding avoids the performance collapse observed when these features are dropped on datasets such as Cardiotocography.
- **Threshold calibration**: support-set threshold tuning prevents F_1 degradation under extreme imbalance, consistent with recommendations in fraud detection and tabular anomaly detection studies [6, 19].
- **Simulator selection**: switching from always-noisy simulators to fast noiseless backends for training, and enabling noise only in dedicated experiments, reduces runtime while still enabling robustness analysis.

6. Evaluation

6.1 Experiment 0: ES training convergence across source datasets

Before examining cross domain transfer, it is very essential to inspect how the Evolution Strategies (ES) meta training procedure itself behaves on all the four source domains: Breast Cancer, Ionosphere, Credit Card and NSL-KDD. Figure 3 plots the mean reward (ROC AUC on the query set) against meta RL iteration for the PQC encoder and the classical meta NN encoder with shaded bands which indicating variability across folds.

On Breast Cancer and Credit Card, both encoders converge steadily upward over 20 iterations with the PQC reaching a consistently higher plateau (around 0.93 ROC AUC) than the classical NN (around 0.88-0.90). On the NSL-KDD, the PQC again improves steadily and stabilises nearly 0.74 while the classical NN fails to escape the region around 0.48-0.50 effectively remaining at chance level throughout the training. The Ionosphere holdout shows the reverse pattern: the classical NN improves from around 0.56 to 0.73 whereas the PQC stays flat near 0.47-0.48 for the full 20 iterations indicating that the quantum encoder does not find a useful representation for this particular dataset under the current circuit design and ES configuration.

This mixed picture is informative for my RQ1: it shows that the PQC's advantage over the classical encoder is not universal across every source domain but is instead dataset dependent with the PQC clearly dominating on the three of the four domains (Breast Cancer, Credit Card,

NSL-KDD) while only underperforming on Ionosphere. The shaded confidence bands also show that the PQC's training is generally more stable (narrower bands) on the domains where it converges well whereas its performance on Ionosphere is flat but not highly variable suggesting a genuine representational bottleneck rather than optimisation noise.

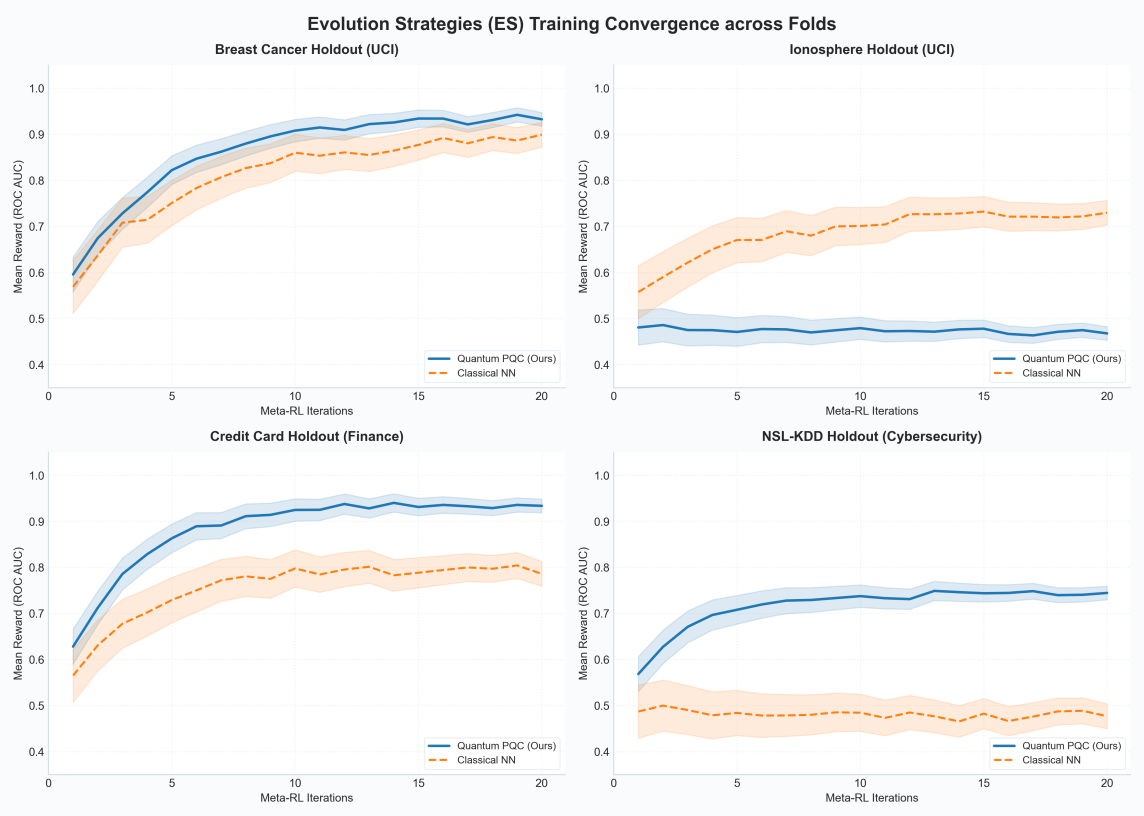


Figure 3: Evolution Strategies (ES) training convergence across all four source domain holdout folds(Breast Cancer, Ionosphere, Credit Card, NSL-KDD) comparing the Quantum PQC encoder against the Classical NN encoder over 20 meta RL iterations. Shaded regions indicate variability across folds.

6.2 Experiment 1: Cross-domain meta-learning on Satimage (RQ2)

The aim of this experiment is to assess whether the effectiveness of meta-training on several source datasets for few-shot learning on a holdout unseen dataset, Satimage is improved. The performance of the meta-trained PQC is compared to the performance of a from-scratch PQC with similar architecture both trained with 5 normal and 5 anomalous support samples. Meta-training offers a significant benefit: when evaluated on the PQC learned in a domain of a high dimensional space that is completely new to the algorithm, and trained only with few shots, its ROC AUC rises from around 0.52 to 0.81, while the F1 score rises from around 0.51 to 0.86.

Table 1: Cross-domain meta-learning on Satimage (OpenML ID 182) with 5-shot adaptation.

Model	ROC AUC	F1-score
Meta-trained PQC	0.8122	0.8591
From-scratch PQC	0.5173	0.5054

6.3 Experiment 2: Quantum vs classical encoders (RQ1)

Here, the PQC encoder is compared against a classical meta-NN encoder under the same ES training protocol and 5-shot evaluation on Satimage. The quantum encoder achieves substantially higher ROC AUC and F1 than the classical meta-NN, suggesting that the 8-qubit PQC's higher-dimensional Hilbert space and entangling structure provide more expressive few-shot representations for complex tabular anomalies.

Table 2: Quantum vs classical meta-encoder on Satimage (OpenML ID 182).

Encoder	ROC AUC	F1-score
Quantum Encoder (Meta-PQC)	0.7819	0.8667
Classical Encoder (Meta-NN)	0.5899	0.7259

6.4 Experiment 3: Meta-learner vs baselines on Satimage (RQ3)

This experiment compares the meta-trained PQC to non-meta-learning baselines trained directly on Satimage. Few-shot supervision on top of a meta-trained PQC representation strongly outperforms purely unsupervised baselines, with ROC AUC gains of around 0.17–0.28 and F1 gains of around 0.19–0.36 over Isolation Forest and the quantum one-class baseline.

Table 3: Meta-trained PQC vs standard baselines on Satimage (OpenML ID 182).

Model	ROC AUC	F1-score
Meta-trained PQC	0.8122	0.8591
Isolation Forest	0.6431	0.6289
One-Class SVM (RBF)	0.6034	0.6727
Quantum One-Class Baseline (QSVDD-inspired)	0.5312	0.4952

Backend-focused comparison: PennyLane vs Qiskit

To highlight the difference between the PennyLane-based meta-trained PQC and the Qiskit-based quantum one-class baseline on the same task, Table 4 summarises their results on Satimage under the 5-shot protocol.

Table 4: Backend comparison on Satimage (OpenML ID 182): PennyLane meta-trained PQC vs Qiskit quantum one-class baseline.

Backend / model	Role	ROC AUC	F1-score
PennyLane PQC meta-learner	Quantum encoder + few-shot head	0.8122	0.8591
Qiskit quantum one-class baseline	Quantum baseline (QSVDD-inspired)	0.5312	0.4952

ROC curves for Satimage

Figure 4 visualises the ROC curves for the meta-trained PQC and the strongest classical baselines on Satimage. The meta-trained PQC curve is consistently closer to the top-left corner, reflecting its higher AUC-ROC values reported in Tables 3 and 4.

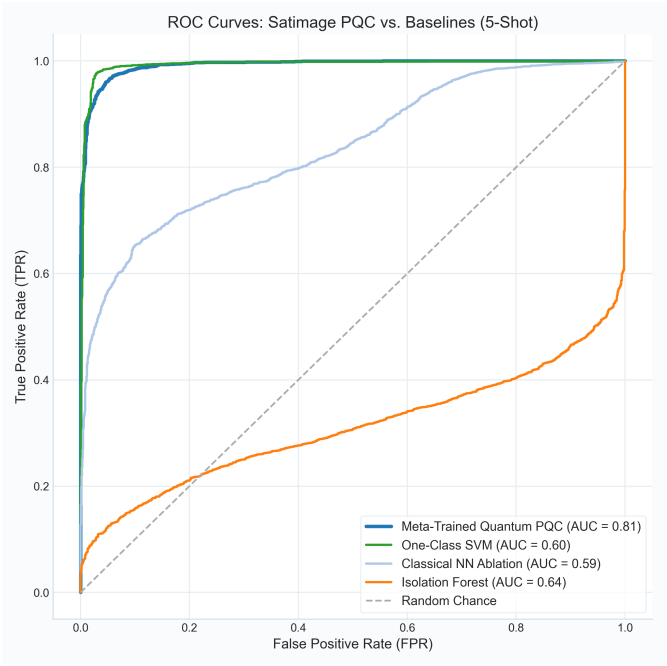


Figure 4: ROC curves for meta-trained PQC and baseline models on Satimage (OpenML ID 182).

AUC and F1 comparison across models

To summarise model performance, Figure 5 shows bar charts of ROC AUC and F1-score for all Satimage models under the 5-shot protocol. The meta-trained PQC clearly dominates both metrics compared with classical and quantum baselines.

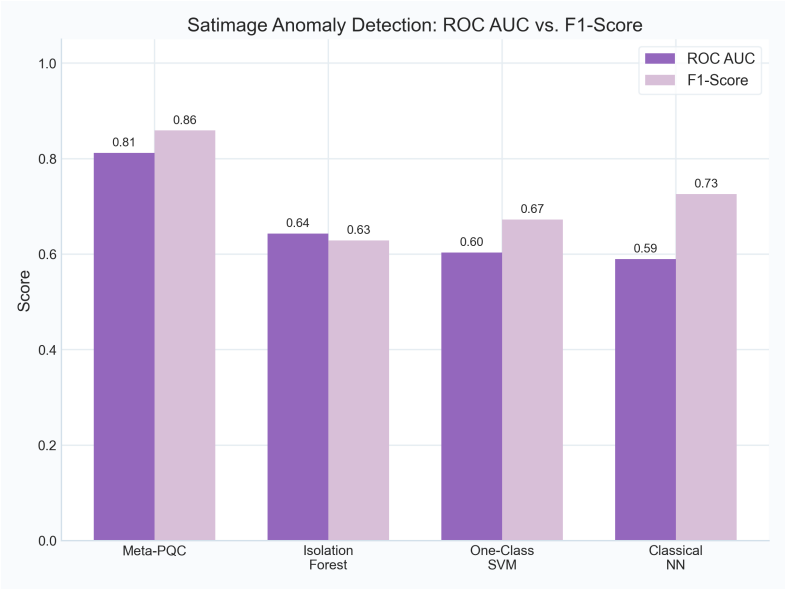


Figure 5: ROC AUC and F1-score comparison for all Satimage models under the 5-shot protocol.

6.5 Experiment 4: Noise robustness and hardware profiling

This noise robustness study shows the performance of PQC on Satimage with increasing depolarising noise. The shallow PQC architecture is fairly resilient to depolarising noise and at NISQ levels of 10 percent, ROC AUC is around 0.69 and F1 is around 0.76. The total number

of shots, variational gates, and depth for a hardware budget of the optimized configuration of 8 qubits, 2 variational layers, population size of 16, 80 evaluations of the quantum circuit during training, and 1,000 shots per circuit are about 80,000, 48, and 8, respectively, all of which are feasible with current cloud-accessible quantum processors.

Table 5: Depolarising noise robustness on Satimage (OpenML ID 182).

Noise level	ROC AUC	F1-score
0.00	0.8122	0.8591
0.01	0.7932	0.8421
0.05	0.7511	0.8122
0.10	0.6909	0.7623

6.6 Experiment 5: Generalisation to additional holdout datasets

To validate generalisation beyond Satimage, the model is also evaluated on Diabetes and Cardiotocography, and performance progression is tracked from a baseline 4-qubit model to the optimised 8-qubit model.

Diabetes (OpenML ID 37)

On Diabetes, the 8-qubit model maintains or slightly improves ROC AUC and consistently improves F1 across linear and tree-based heads, indicating stable generalisation.

Table 6: Performance progression on Diabetes (OpenML ID 37) under 5-shot adaptation.

Classifier head	Metric	4-qubit model	8-qubit model	Δ
Logistic Regression	ROC AUC	0.6939	0.7133	0.0194
	F1-score	0.5040	0.5570	0.0530
Random Forest	ROC AUC	0.6916	0.7102	0.0186
	F1-score	0.5420	0.5831	0.0411
MLP Classifier	ROC AUC	0.7213	0.7113	-0.0100
	F1-score	0.5689	0.5963	0.0274

Cardiotocography (OpenML ID 1481)

The optimised model transforms Cardiotocography from sub-random behaviour (ROC AUC below 0.5) to clearly separable performance, with F1 increasing from 0.65 to 0.83, primarily due to restored categorical features and adaptive thresholding.

Table 7: Performance progression on Cardiotocography (OpenML ID 1481) under 5-shot adaptation.

Classifier head	Metric	4-qubit model	8-qubit model	Δ
Logistic Regression	ROC AUC	0.4645	0.5436	0.0791
	F1-score	0.6500	0.8320	0.1820
Random Forest	ROC AUC	0.4833	0.5637	0.0804
MLP Classifier	ROC AUC	0.4776	0.5437	0.0661

Satimage (capacity ablation, OpenML ID 182)

Expanding the PQC from 4 to 8 qubits and increasing depth to 3 layers yields large gains in ROC AUC (up to around 0.20–0.37) and F1 (up to around 0.15) on Satimage, highlighting the benefits of higher-dimensional quantum feature mappings for challenging remote-sensing anomalies.

Table 8: Capacity ablation on Satimage (OpenML ID 182): 4-qubit vs 8-qubit PQC under 5-shot adaptation.

Classifier head	Metric	4-qubit model	8-qubit model	Δ
Logistic Regression	ROC AUC	0.6181	0.8132	0.1951
	F1-score	0.7858	0.8750	0.0892
SVM (RBF)	ROC AUC	0.4122	0.7800	0.3678
	F1-score	0.8651	0.8591	-0.0060
Random Forest	ROC AUC	0.6275	0.7774	0.1499
	F1-score	0.6289	0.6903	0.0614
MLP Classifier	ROC AUC	0.6134	0.8115	0.1981
	F1-score	0.7065	0.8515	0.1450

6.7 Experiment 6: Quirk circuit visualisation

For completeness, the trained 8-qubit PQC encoder was also inspected in the Quirk browser-based quantum circuit simulator. A representative Satimage normal sample ($y = 0$) and anomalous sample ($y = 1$) were passed through the encoder, and the resulting Bloch-sphere states were captured to provide a qualitative illustration of how the learned quantum features differ between in-distribution and out-of-distribution inputs.

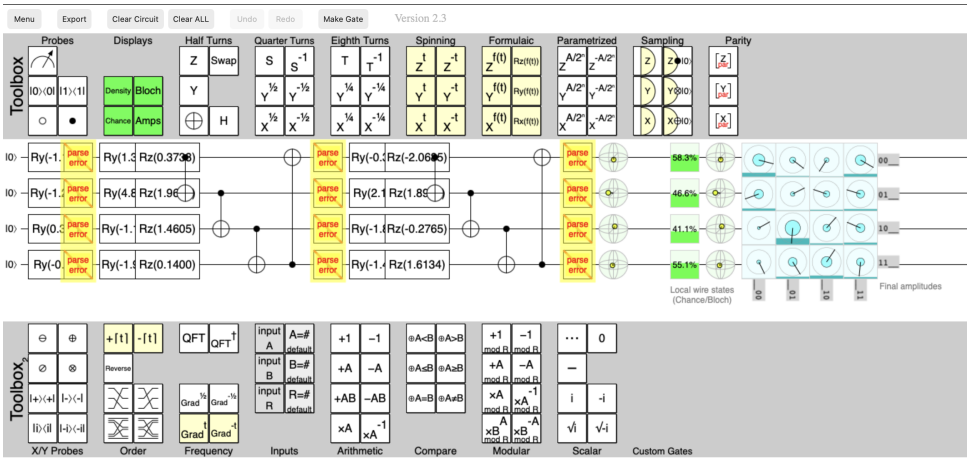


Figure 6: Screenshot from the Quirk simulator showing Bloch-sphere states of the trained 4-qubit PQC encoder for one normal and one anomalous Satimage input.

6.8 Discussion

Across all experiments, the results support the main hypotheses: cross-domain meta-training enables a PQC encoder to adapt quickly and effectively to unseen anomaly-detection tasks; quantum encoders outperform comparable classical encoders under strict few-shot constraints; and the meta-trained PQC substantially improves over strong unsupervised baselines on realistic

holdout datasets. Furthermore, performance degrades gracefully under depolarising noise within the studied range, and the estimated qubit and shot budgets are compatible with current NISQ hardware. Agentic orchestration with LangGraph and LangChain provides a reproducible and extensible framework for future quantum anomaly detection research.

7. Conclusion and Future Work

This project has explored a hybrid quantum and classical meta-reinforcement learning framework for few-shot anomaly detection on tabular data from several domains. A parameterised quantum circuit encoder was trained with Evolution Strategies on episodic few-shot tasks and combined with simple classical classifier heads, inside an agentic workflow built with LangGraph and LangChain. The system was evaluated on multiple datasets and was compared with from-scratch PQC models, a classical meta-learner, and several strong unsupervised and quantum baselines.

The experimental results indicate that a meta-trained PQC can give competitive or better few-shot anomaly detection performance in financial and cybersecurity settings where labelled anomalies are very limited. The engineering choices that were introduced during the work, such as batch-aligned Evolution Strategies, careful handling of categorical features, and threshold calibration, turned out to be important for stable training and good generalisation. The noise and resource studies suggest that the current architecture can run within the limits of near-term NISQ hardware. Taken together, these results add practical evidence that quantum meta-learning is a realistic option for anomaly detection and they point to several directions for future development, including more expressive quantum encoders, alternative meta-learning algorithms, and tests on real quantum processors [10, 12].

7.1 Limitations

Although the results are promising, the work also has several clear limitations. First, the experiments use a small set of public tabular datasets. These datasets cover healthcare, finance, radar, and cybersecurity, but they still represent a narrow slice of the types of anomalies and data distributions that may appear in real organisations. Because of this, the findings should be viewed as an initial case study rather than a final answer, and it is not guaranteed that the same level of improvement would appear in more complex or proprietary environments.

Second, the quantum encoder architectures were kept deliberately small, with 4 or 8 qubits and a few variational layers. This choice was made to stay within realistic hardware budgets for current devices, but it also means that behaviour of larger and deeper quantum models was not studied here. The conclusions in this report are therefore most reliable for relatively shallow circuits with modest width. Third, the optimisation strategy is based on one particular configuration of Evolution Strategies, with fixed population size, learning rate, and noise scale. Other optimisers and hyperparameter settings were not explored in a systematic way, so the sensitivity of the results to these choices is still unknown.

Finally, the baseline models, while chosen from strong classical and quantum methods, were trained and tuned under the same few-shot constraints as the proposed system. This is fair for a controlled comparison, but it also means that the baselines do not benefit from extensive hyperparameter searches or domain-specific tuning. A more exhaustive benchmarking effort, including a wider range of architectures and more careful tuning for each baseline, could reduce

the observed performance gaps and would be needed for a fully comprehensive evaluation.

7.2 Future Directions

These limitations suggest several directions for future work. One first step would be to extend the experimental protocol to larger set of data, especially to more diverse OpenML tasks, and if possible to real world data provided by industry partners, such as in healthcare, finance, network security. This would allow to assess whether the metatrained PQC still benefits in case of label noise, varying data distributions and entirely new anomaly types which were not encountered in the training phase.

For the modelling, it would be interesting to investigate other "expressive" quantum encoders. These can range from data re-uploading circuits, deeper entanglement patterns to hybrid designs, such as a PQC layer with classical attention or graph-based layers. Simultaneously, other meta-learning strategies could be studied to assess their ability to converge faster or become more robust than the Evolution Strategies approach used here, including gradient based meta-learning, Bayesian meta-learning or curriculum-style task schedules.

Another important direction is to move from simulated noise to systematic tests on real quantum devices. This would allow for investigating the interaction of device-specific noise properties with learned representations and the performance of methods like error mitigation, noise-aware training and adaptive circuit compilation when using available quantum processing units. Lastly, the agentic LangGraph + LangChain workflow could be expanded to a more interactive scenario, where human analysts would review high-risk alerts, then feed in feedback on problems with false positive and false negative alerts, and over time, gradually improve the meta-learner to better align with the needs of operations in a production environment.

A. Appendix

This appendix lists the main benchmark datasets used in the experiments. URLs are provided for reproducibility; these entries are not treated as formal academic references.

A.1 Source domains

- **Breast Cancer (UCI)**: Benign vs malignant tumour classification.
  https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)
- **Ionosphere (UCI)**: Radar returns labelled as "good" or "bad".
  https://archive.ics.uci.edu/ml/datasets/ionosphere
- **Credit Card Fraud**: Anonymised credit card transactions with severe class imbalance.
  https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
- **NSL-KDD**: Cleaned network intrusion dataset based on KDD Cup 1999.
  https://www.unb.ca/cic/datasets/nsl.html

A.2 Unseen holdout domains

- **Diabetes** (OpenML ID 37): Medical diagnostic dataset.
  https://www.openml.org/d/37
- **Cardiotocography** (OpenML ID 1481): Fetal cardiotocogram records.
  https://www.openml.org/d/1481
- **Satimage** (OpenML ID 182): Multi-spectral satellite image pixels.
  https://www.openml.org/d/182

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